

New Keynesian Policy Design

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1 Efficient allocation and distortions

Monopolistic competition + flexible prices $\Rightarrow$ efficient allocation, with a subsidy to correct distortions of monopolistic competition.

Sticky prices + policy that fully stabilizes the price level $\Rightarrow$ efficient allocation.

2 Efficient allocation under social planner's problem

$$\begin{aligned} & \max_{\{C_t, N_t\}} U(C_t, N_t) \\ \text{s.t.} \quad & C_t(i) = A_t F(N_t(i)), \quad \text{for all } i \in [0, 1], \\ & C_t = \left(\int_0^1 C_t(i)^{\frac{\varepsilon-1}{\varepsilon}} di \right)^{\frac{\varepsilon}{\varepsilon-1}}, \quad N_t = \int_0^1 N_t(i) di. \\ & C_t(i) = C_t, \quad \forall i \in [0, 1], \quad (1) \\ & N_t(i) = N_t, \quad \forall i \in [0, 1], \quad (2) \\ & -\frac{U_{N,t}}{U_{C,t}} = MPN_t = (1 - \alpha) A_t N_t^{-\alpha}, \quad (3) \end{aligned}$$

$$\text{nominal rigidity} \Rightarrow \begin{cases} C_t(i) \neq C_t, \\ N_t(i) \neq N_t, \end{cases}$$

$$\text{nominal rigidity or monopolistic competition} \Rightarrow (3) \text{ does not hold.}$$

$$(1)\&(2) \text{ symmetric allocation} \Rightarrow MPN_t = MPN_t(i).$$

Optimality conditions can be met due to:

i. All goods enter utility symmetrically.

For planner, weight are the same even distribution $C_t(i)$.

ii. Identical technology of production.

For planner, technology for i is the same, even distribution $N_t(i)$.

iii. Concavity of utility.

For planner, there is an optimal allocation of $C_t(i)$ and $N_t(i)$.

$$\text{MRS}_{C,N} = -\frac{U_{N,t}}{U_{C,t}}, \quad \text{MRT}_{C,N} = \frac{dC_t}{dN_t} = F_N(N_t) A_t \quad \text{output transferred one-for-one to consumption.}$$

$$\Rightarrow \text{MRS}_{C,N} = \text{MRT}_{C,N} = \text{MPN}.$$

3 Suboptimality in NK

Two sources of distortion:

- 1) Monopolistic competitive firms $\Rightarrow$ market power.
- 2) Sticky price.

Let's study (1) separately. Assume flexible price. With isoelastic demand function, optimal pricing:

$$P_t = \mu \frac{W_t}{\text{MPN}_t}, \quad \mu = \frac{\varepsilon}{\varepsilon - 1} \quad \text{markup.}$$

$$\Rightarrow \text{MRS}_{C,N} = -\frac{U_{N,t}}{U_{C,t}} = \frac{W_t}{P_t} = \frac{\text{MPN}_t}{\mu}.$$

$\mu > 1 \Rightarrow$ low employment and output, and optimal condition (3) is violated.

Let τ denotes the employment subsidy; subsidy financed by lump-sum tax, **no distortions**.

$$P_t = \mu \frac{(1 - \tau) W_t}{\text{MPN}_t} \Rightarrow -\frac{U_{N,t}}{U_{C,t}} = \frac{W_t}{P_t} = \frac{\text{MPN}_t}{\mu (1 - \tau)}.$$

$$\text{MPN}_t = \frac{\text{MPN}_t}{\mu (1 - \tau)} \Rightarrow \mu (1 - \tau) = 1 \Rightarrow \tau = \frac{1}{\varepsilon}.$$

The optimal allocation can be achieved by giving employment subsidy of $\tau = \frac{1}{\varepsilon}$.

Now study (2) price stickiness.

3.1 Average markup varies over time

$$\mu_t = \frac{P_t}{(1 - \tau) W_t / \text{MPN}_t} = \frac{\mu P_t}{W_t / \text{MPN}_t} \Rightarrow \frac{W_t}{P_t} = \frac{\mu}{\mu_t} \text{MPN}_t.$$

$$-\frac{U_{N,t}}{U_{C,t}} = \frac{W_t}{P_t} = \frac{\mu}{\mu_t} \text{MPN}_t.$$

If $\mu_t \neq \mu$, optimal allocation condition (3) is violated.

3.2 Staggered price setting: sticky price

$$P_t(i) \neq P_t(j), \quad C_t(i) \neq C_t(j), \quad N_t(i) \neq N_t(j).$$

Optimal conditions (1) and (2) are violated.

4 Optimal monetary policy in the basic NK

Assumptions:

1. An optimal subsidy to offset distortion of firms' market power.
2. Economy starts without relative price distortion:

$$P_{-1}(i) = P_{-1}, \quad \text{for all } i \in [0, 1].$$

Optimal policy should achieve:

1. $P_t = P_{t-1}, \forall t = 0, 1, 2, \dots$, so that conditions (1) and (2) hold.
2. $\mu_t = \mu, \forall t$, so that condition (3) holds.

$$\Leftrightarrow \tilde{y}_t = \hat{y}_t - \hat{y}_t^n = 0, \quad \forall t; \quad \hat{\pi}_t = 0, \quad \forall t.$$

DIS:

$$\begin{aligned} \tilde{y}_t &= -\frac{1}{\sigma} \left(\hat{i}_t - \mathbb{E}_t \{ \hat{\pi}_{t+1} \} - \hat{r}_t^n \right) + \mathbb{E}_t \{ \tilde{y}_{t+1} \}. \\ &\Rightarrow \hat{i}_t = \hat{r}_t^n, \quad \forall t. \end{aligned}$$

1. **Stabilizing output:** Not freezing output at constant.

$$\hat{y}_t = \hat{y}_t^n, \quad \forall t.$$

Eliminating gap, not eliminating fluctuations.

$\hat{y}_t^n$ is subject to technology shock.

2. **Stabilizing price:** divine coincidence.

If central bank achieves price stability, then it automatically gets $\tilde{y}_t = 0$.

Stabilizing inflation $\Rightarrow$ stabilizing output gap.

4.1 How to implement the optimal policy?

Non-policy block:

$$\text{DIS: } \tilde{y}_t = -\frac{1}{\sigma} \left(\hat{i}_t - \mathbb{E}_t \{ \hat{\pi}_{t+1} \} - \hat{r}_t^n \right) + \mathbb{E}_t \{ \tilde{y}_{t+1} \},$$

$$\text{NKPC: } \hat{\pi}_t = \beta \mathbb{E}_t \{ \hat{\pi}_{t+1} \} + \kappa \tilde{y}_t.$$

When $\hat{\pi}_t = 0, \forall t \Rightarrow \tilde{y}_t = 0, \forall t$. divine coincidence.

Policy block:

Case a: An exogenous interest rate rule

$$\hat{i}_t = \hat{r}_t^n, \quad \forall t.$$

$$\Rightarrow \begin{cases} \boxed{\tilde{y}_t} = \frac{1}{\sigma} \mathbb{E}_t \{ \hat{\pi}_{t+1} \} + \mathbb{E}_t \{ \tilde{y}_{t+1} \}, \\ \hat{\pi}_t = \beta \mathbb{E}_t \{ \hat{\pi}_{t+1} \} + \kappa \tilde{y}_t \end{cases}$$

Substitute the first equation into the second:

$$\begin{aligned}\hat{\pi}_t &= \beta E_t \{ \hat{\pi}_{t+1} \} + \kappa \left(\frac{1}{\sigma} E_t \{ \hat{\pi}_{t+1} \} + E_t \{ \tilde{y}_{t+1} \} \right) \\ &= \kappa E_t \{ \tilde{y}_{t+1} \} + \left(\beta + \frac{\kappa}{\sigma} \right) E_t \{ \hat{\pi}_{t+1} \} . \\ \Rightarrow \begin{bmatrix} \tilde{y}_t \\ \hat{\pi}_t \end{bmatrix} &= A_0 \begin{bmatrix} E_t \{ \tilde{y}_{t+1} \} \\ E_t \{ \hat{\pi}_{t+1} \} \end{bmatrix}, \quad \text{where } A_0 = \begin{bmatrix} 1 & \frac{1}{\sigma} \\ \kappa & \beta + \frac{\kappa}{\sigma} \end{bmatrix} .\end{aligned}$$

$\tilde{y}_t = \hat{\pi}_t = 0$ is one solution associated with optimal policy, but not unique.

$\tilde{y}_t$ and $\hat{\pi}_t$ are neither predetermined.

$$\det(A_0) = \beta, \quad \text{tr}(A_0) = 1 + \beta + \frac{\kappa}{\sigma}.$$

$$P(1) = 1 - T + D = 1 - 1 - \beta - \frac{\kappa}{\sigma} + \beta = -\frac{\kappa}{\sigma} < 0.$$

$$P(-1) = 1 + T + D = 1 + 1 + \beta + \frac{\kappa}{\sigma} + \beta = 2 + 2\beta + \frac{\kappa}{\sigma} > 0.$$

There is one root between $(-1, 1)$ and another one outside the unit circle.

jump variables > # unstable roots $\Rightarrow$ system is indeterminate.

No guarantee on the realization of $\tilde{y}_t = \hat{\pi}_t = 0, \quad \forall t$.

Case b: A Taylor-type interest rate rule

$$\hat{i}_t = \hat{r}_t^n + \phi_\pi \hat{\pi}_t + \phi_y \tilde{y}_t, \quad \phi_\pi \geq 0, \quad \phi_y \geq 0, \quad \forall t.$$

$$\begin{cases} \tilde{y}_t = -\frac{\phi_\pi}{\sigma} \hat{\pi}_t - \frac{\phi_y}{\sigma} \tilde{y}_t + \frac{1}{\sigma} E_t \{ \hat{\pi}_{t+1} \} + E_t \{ \tilde{y}_{t+1} \}, \\ \hat{\pi}_t = \beta E_t \{ \hat{\pi}_{t+1} \} + \kappa \tilde{y}_t. \end{cases}$$

$$\begin{cases} \tilde{y}_t = \frac{1 - \beta\phi_\pi}{\sigma + \phi_y + \kappa\phi_\pi} E_t \{ \hat{\pi}_{t+1} \} + \frac{\sigma}{\sigma + \phi_y + \kappa\phi_\pi} E_t \{ \tilde{y}_{t+1} \}, \\ \hat{\pi}_t = \frac{\kappa + \beta\sigma + \beta\phi_y}{\sigma + \phi_y + \kappa\phi_\pi} E_t \{ \hat{\pi}_{t+1} \} + \frac{\sigma\kappa}{\sigma + \phi_y + \kappa\phi_\pi} E_t \{ \tilde{y}_{t+1} \}. \end{cases}$$

$$\begin{bmatrix} \tilde{y}_t \\ \hat{\pi}_t \end{bmatrix} = A_T \begin{bmatrix} E_t \{ \tilde{y}_{t+1} \} \\ E_t \{ \hat{\pi}_{t+1} \} \end{bmatrix},$$

$$A_T = \Omega \begin{bmatrix} \sigma & 1 - \beta\phi_\pi \\ \kappa\sigma & \kappa + \beta(\sigma + \phi_y) \end{bmatrix}, \quad \Omega = \frac{1}{\sigma + \phi_y + \kappa\phi_\pi}.$$

$$\tilde{y}_t = \hat{\pi}_t = 0, \quad \forall t \quad \text{is always a solution.}$$

To guarantee uniqueness, we need two stable roots

notice that $x_t = A_T E_t x_{t+1}$, but BK is defined on $E_t x_{t+1} = A_T^{-1} x_t$,

2 stable roots in $A_T \iff$ 2 unstable roots in A_T^{-1} .

$$\text{tr}(A_T) = \Omega [\sigma + \kappa + \beta (\sigma + \phi_y)] , \quad \det(A_T) = \Omega \beta \sigma.$$

To have stable result, need $D < 1$, $P(1) > 0$, $P(-1) > 0$.

$$\begin{aligned} P(1) &= 1 - \text{tr}(A_T) + \det(A_T) = 1 - \Omega [\sigma + \kappa + \beta \phi_y] > 0 \\ &\Rightarrow \sigma + \kappa + \beta \phi_y < \sigma + \phi_y + \kappa \phi_\pi \\ &\Rightarrow \kappa (\phi_\pi - 1) + (1 - \beta) \phi_y > 0 \quad (*) \end{aligned}$$

$$P(-1) = 1 + \text{tr}(A_T) + \det(A_T) = 1 + \Omega [\sigma + \kappa + \beta \phi_y + \beta \sigma] > 0 \quad \text{for sure.}$$

$$\Delta = \text{Tr}(A_T)^2 - 4 \det(A_T) = \Omega^2 [\sigma + \kappa + \beta (\sigma + \phi_y)]^2 - 4 \Omega \beta \sigma, \quad \text{sign does not matter.}$$

$$\begin{aligned} D = \det(A_T) &= \Omega \beta \sigma = \frac{\beta \sigma}{\sigma + \phi_y + \kappa \phi_\pi} = \frac{1}{\frac{1}{\beta} + \frac{\phi_y + \kappa \phi_\pi}{\beta \sigma}} \\ &< \frac{1}{1 + \frac{\kappa - \kappa \phi_\pi + \kappa \phi_\pi}{\beta \sigma}} = \frac{1}{1 + \frac{\kappa}{\beta \sigma}} < 1. \end{aligned}$$

If $\kappa (\phi_\pi - 1) + (1 - \beta) \phi_y > 0$ is satisfied, $\tilde{y}_t = \hat{\pi}_t = 0$ and $\hat{i}_t = \hat{r}_t^n$ hold $\forall t$.

If $\phi_y = 0 \Rightarrow \kappa (\phi_\pi - 1) > 0 \Rightarrow \phi_\pi > 1$ Taylor principle.

If $\phi_y > 0$, output gap helps, lower bound on ϕ_π can be relaxed a little.

But $1 - \beta$ is usually small, inflation is the main stabilizing force.

Case c: A forward-looking interest rate rule

$$\hat{i}_t = \hat{r}_t^n + \phi_\pi \mathbb{E}_t \{\hat{\pi}_{t+1}\} + \phi_y \mathbb{E}_t \{\tilde{y}_{t+1}\}.$$

$$\begin{cases} \tilde{y}_t = \frac{1 - \phi_\pi}{\sigma} \mathbb{E}_t \{\hat{\pi}_{t+1}\} + \frac{\sigma - \phi_y}{\sigma} \mathbb{E}_t \{\tilde{y}_{t+1}\}, \\ \hat{\pi}_t = \beta \mathbb{E}_t \{\hat{\pi}_{t+1}\} + \kappa \tilde{y}_t. \end{cases} \quad (*)$$

$$\begin{cases} \tilde{y}_t = \frac{1 - \phi_\pi}{\sigma} \mathbb{E}_t \{\hat{\pi}_{t+1}\} + \frac{\sigma - \phi_y}{\sigma} \mathbb{E}_t \{\tilde{y}_{t+1}\}, & (**) \\ \hat{\pi}_t = \frac{\kappa (1 - \phi_\pi) + \beta \sigma}{\sigma} \mathbb{E}_t \{\hat{\pi}_{t+1}\} + \frac{\kappa (\sigma - \phi_y)}{\sigma} \mathbb{E}_t \{\tilde{y}_{t+1}\}. & (***) \end{cases}$$

$$\begin{bmatrix} \tilde{y}_t \\ \hat{\pi}_t \end{bmatrix} = A_F \begin{bmatrix} \mathbb{E}_t \{\tilde{y}_{t+1}\} \\ \mathbb{E}_t \{\hat{\pi}_{t+1}\} \end{bmatrix},$$

$$A_F = \begin{bmatrix} \frac{\sigma - \phi_y}{\sigma} & \frac{1 - \phi_\pi}{\sigma} \\ \frac{\kappa (\sigma - \phi_y)}{\sigma} & \beta + \frac{\kappa (1 - \phi_\pi)}{\sigma} \end{bmatrix}.$$

$$\text{tr}(A_F) = \beta + \frac{\kappa (1 - \phi_\pi) + \sigma - \phi_y}{\sigma}, \quad \det(A_F) = \beta \frac{\sigma - \phi_y}{\sigma}.$$

$$\begin{aligned}
P(1) &= 1 - \text{tr}(A_F) + \det(A_F) \\
&= 1 - \beta - \frac{\kappa(1 - \phi_\pi) + (\sigma - \phi_y)}{\sigma} + \beta \frac{\sigma - \phi_y}{\sigma} > 0 \\
&\Rightarrow \kappa(\phi_\pi - 1) + (1 - \beta)\phi_y > 0 \quad (1) \quad \text{policy cannot be too passive.}
\end{aligned}$$

$$\begin{aligned}
P(-1) &= 1 + \text{tr}(A_F) + \det(A_F) \\
&= 1 + \beta + \frac{\kappa}{\sigma}(1 - \phi_\pi) + 1 - \frac{1}{\sigma}\phi_y + \beta - \frac{\beta}{\sigma}\phi_y > 0 \\
&\Rightarrow \kappa(\phi_\pi - 1) + (1 + \beta)\phi_y < 2\sigma(1 + \beta) \quad (2) \quad \text{policy cannot be too aggressive.}
\end{aligned}$$

$$D < 1 \Rightarrow \beta \left(1 - \frac{\phi_y}{\sigma}\right) < 1, \quad \text{since } \beta \in (0, 1) \text{ and } \phi_y \geq 0, \text{ so this holds.}$$

$\Rightarrow$ Under (1) and (2), we have 2 stable roots in A_F , 2 unstable roots in A_F^{-1} .

$$\tilde{y}_t = \hat{\pi}_t = 0, \quad \forall t, \quad \text{is the unique solution.}$$

1) If $\phi_\pi > 1$, the coefficient on $\mathbb{E}_t \{\hat{\pi}_{t+1}\}$ is negative.

$$\mathbb{E}_t \{\hat{\pi}_{t+1}\} \uparrow \Rightarrow \tilde{y}_t \downarrow \Rightarrow \hat{\pi}_t \downarrow.$$

If $\phi_\pi < 1$, then

$$\mathbb{E}_t \{\hat{\pi}_{t+1}\} \uparrow \Rightarrow \tilde{y}_t \uparrow \Rightarrow \hat{\pi}_t \uparrow,$$

self-fulfilling inflation; policy cannot be too passive.

2) If policy is too aggressive so that condition (2) is violated,

$$P(-1) < 0, \quad \text{one eigenvalue crosses } -1, \quad \text{system unstable.}$$

But all optimal policy listed are not realistic, since policy rate must be adjusted one-for-one with the natural interest rate.

But $\hat{r}_t^n$ is usually unobservable, because it depends on the true model of the economy,

parameter values and the realized shocks. $\Rightarrow$ Simple rules are needed!

Due to divine coincidence, unobservable output gap is not the main issue, since $E_t \hat{\pi}_{t+1} = 0 \Rightarrow \tilde{y}_t = 0$.

$$\kappa(\phi_\pi - 1) + (1 - \beta)\phi_y = 0 \quad \Longleftrightarrow \quad \phi_\pi = 1 - \frac{1 - \beta}{\kappa}\phi_y.$$

$$\kappa(\phi_\pi - 1) + (1 + \beta)\phi_y = 2\sigma(1 + \beta) \quad \Longleftrightarrow \quad \phi_\pi = 1 + \frac{2\sigma(1 + \beta)}{\kappa} - \frac{1 + \beta}{\kappa}\phi_y.$$

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Figure 1: Determinacy region

5 Simple monetary policy rule

5.1 Rotemberg and Woodford (1999)

Second-order approximation to the utility losses due to deviations from efficient allocation:

$$W = \frac{1}{2} \mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t \left[\left(\sigma + \frac{\varphi + \alpha}{1 - \alpha} \right) \tilde{y}_t^2 + \frac{\varepsilon}{\kappa} \hat{\pi}_t^2 \right].$$

Per-period loss:

$$L = \frac{1}{2} \left[\left(\sigma + \frac{\varphi + \alpha}{1 - \alpha} \right) \text{Var}(\tilde{y}_t) + \frac{\varepsilon}{\kappa} \text{Var}(\hat{\pi}_t) \right].$$

$\sigma \uparrow, \varphi \uparrow, \alpha \uparrow \Rightarrow L \uparrow$. Curvature amplifies the effect (consumption, labor, technology).

$\varepsilon \uparrow, \frac{1}{\kappa} \uparrow \Rightarrow L \uparrow$. Amplify losses from price dispersion; $\frac{1}{\lambda}$ amplifies the degree of price dispersion.

Lecture 9 Supplements: Policy Rules and Welfare Loss

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Rotemberg and Woodford (1999): second-order approximation to household utility around efficient allocation

$$\tilde{y}_t = y_t - y_t^n, \quad Y_t = A_t N_t^{1-\alpha} \Rightarrow \tilde{y}_t = (1-\alpha) \tilde{n}_t \Rightarrow \tilde{n}_t = \frac{\tilde{y}_t}{1-\alpha}.$$

$$\text{expand utility around } x = \tilde{y}_t = \tilde{c}_t, \quad U = \frac{C^{1-\delta}}{1-\delta} - \frac{N^{1+\psi}}{1+\psi}.$$

consumption moves by x , labor moves by $x/(1-\alpha)$.

Consumption part: move by x

$$\frac{(C^n e^x)^{1-\sigma} - (C^n)^{1-\sigma}}{(1-\sigma) U_c^n C^n} \quad \text{where } U_c^n = (C^n)^{-\sigma} \quad (*)$$

$$\left. \begin{aligned} (*) &= \frac{e^{(1-\sigma)x} - 1}{1-\sigma}, \\ e^{(1-\sigma)x} &\simeq 1 + (1-\sigma)x + \frac{1}{2}(1-\sigma)^2 x^2 \end{aligned} \right\} \Rightarrow (*) = x + \frac{1}{2}(1-\sigma)x^2.$$

Taylor expansion around 0.

Labor part: Since consumption move by x , labor move by $x/(1-\alpha)$.

Expand labor by $\frac{x}{1-\alpha}$ gives the normalized labor utility change:

$$-\frac{(N^n e^{x/(1-\alpha)})^{1+\psi} - (N^n)^{1+\psi}}{(1+\psi) U_c^n C^n} \quad (**)$$

$$\begin{aligned} \text{Since } -\frac{U_N}{U_C} &= MPN = (1-\alpha) \frac{Y^n}{N^n}, & U_N &= (N^n)^\psi, \\ \Rightarrow \frac{(N^n)^\psi}{U_c^n} &= (1-\alpha) \frac{C^n}{N^n}, & \Rightarrow \frac{(N^n)^{1+\psi}}{U_c^n C^n} &= 1-\alpha. \end{aligned}$$

$$\begin{aligned} (**) &= -\frac{e^{(1+\psi)x/(1-\alpha)} - 1}{1+\psi} \frac{(N^n)^{1+\psi}}{U_c^n C^n} \\ &= -(1-\alpha) \frac{e^{(1+\psi)x/(1-\alpha)} - 1}{1+\psi}. \end{aligned}$$

$$e^{(1+\psi)x/(1-\alpha)} \simeq 1 + \frac{1+\psi}{1-\alpha}x + \frac{1}{2}\left(\frac{1+\psi}{1-\alpha}\right)^2 x^2, \quad \text{expand at } x = 0.$$

$$(**) = -\frac{1-\alpha}{1+\psi} \left(\frac{1+\psi}{1-\alpha}x + \frac{1}{2}\left(\frac{1+\psi}{1-\alpha}\right)^2 x^2 \right) = -x - \frac{1}{2} * \frac{1+\psi}{1-\alpha} x^2.$$

Combine consumption part with labor part:

$$\begin{aligned} x + \frac{1}{2}(1-\sigma)x^2 - x - \frac{1}{2}\frac{1+\psi}{1-\alpha}x^2 \\ = \frac{1}{2}\left(1-\sigma - \frac{1+\psi}{1-\alpha}\right)x^2 = -\frac{1}{2}\left(\sigma - 1 + \frac{1+\psi}{1-\alpha}\right)x^2 = -\frac{1}{2}\left(\sigma + \frac{\alpha+\psi}{1-\alpha}\right)\tilde{y}_t^2, \quad (x = \tilde{y}). \end{aligned}$$

Now derive the inflation part: $P_t(i) \neq P_t(j)$; inflation creates relative-price dispersion.

Cross sectional variance of log relative price $\Rightarrow V = \int_0^1 (p(i) - p)^2 di = \int_0^1 s(i)^2 di$,

$$s(i) = p(i) - p.$$

$$\begin{aligned} Y(i) &= \left(\frac{P(i)}{P}\right)^{-\varepsilon} Y \Rightarrow \log Y(i) = -\varepsilon \log P(i) + \varepsilon \log P + \log Y, \\ \Rightarrow y(i) - y &= -\varepsilon (p(i) - p) = -\varepsilon s(i). \end{aligned}$$

$$\begin{aligned} Y(i) &= AN(i)^{1-\alpha} \Rightarrow N(i) = \left(\frac{Y(i)}{A}\right)^{\frac{1}{1-\alpha}} \Bigg\} \Rightarrow N(i) = \left(\frac{Y}{A}\right)^{\frac{1}{1-\alpha}} e^{-\frac{\varepsilon}{1-\alpha}s(i)}. \\ Y(i) &= Ye^{-\varepsilon s(i)} \end{aligned}$$

$$N = \int_0^1 N(i) di = \left(\frac{Y}{A}\right)^{\frac{1}{1-\alpha}} \int_0^1 e^{-\frac{\varepsilon}{1-\alpha}s(i)} di, \quad \text{ithout price dispersion } s(i) = 0. \quad (1)$$

$$\Rightarrow N^{sym} = \left(\frac{Y}{A}\right)^{\frac{1}{1-\alpha}} \dots \text{symmetrical labor/price.} \quad (2)$$

$$\frac{(1)}{(2)} = \frac{N}{N^{sym}} = \int_0^1 e^{-\frac{\varepsilon}{1-\alpha}s(i)} di, \quad \text{let } a = \frac{\varepsilon}{1-\alpha}.$$

$$e^{-as(i)} \simeq 1 - as(i) + \frac{a^2}{2}s(i)^2, \quad \text{expand around } s(i) = 0.$$

$$\Rightarrow \frac{N}{N^{sym}} = 1 - a \int_0^1 s(i) di + \frac{a^2}{2} \int_0^1 s(i)^2 di. \quad (***)$$

$$\begin{aligned} P &= \left[\int_0^1 P(i)^{1-\varepsilon} di \right]^{\frac{1}{1-\varepsilon}} \\ \Rightarrow 1 &= \left[\int_0^1 \left(\frac{P(i)}{P}\right)^{1-\varepsilon} di \right]^{\frac{1}{1-\varepsilon}} \\ \Rightarrow 1 &= \int_0^1 e^{(1-\varepsilon)s(i)} di, \quad \frac{P(i)}{P} = e^{p(i)-p} = e^{s(i)}. \end{aligned}$$

$$e^{(1-\varepsilon)s(i)} \approx 1 + (1-\varepsilon)s(i) + \frac{1}{2}(1-\varepsilon)^2 s(i)^2.$$

$$\begin{aligned}
1 &= \int_0^1 \left[1 + (1 - \varepsilon) s(i) + \frac{1}{2} (1 - \varepsilon)^2 s(i)^2 \right] di \\
&= 1 + (1 - \varepsilon) \int_0^1 s(i) di + \frac{1}{2} (1 - \varepsilon)^2 \int_0^1 s(i)^2 di. \\
\Rightarrow (1 - \varepsilon) \int_0^1 s(i) di + \frac{1}{2} (1 - \varepsilon)^2 V &= 0, \quad \text{where } V \equiv \int_0^1 s(i)^2 di. \\
\Rightarrow \int_0^1 s(i) di + \frac{1}{2} (1 - \varepsilon) V &= 0 \Rightarrow \int_0^1 s(i) di = \frac{\varepsilon - 1}{2} V. \\
(***) = \frac{N}{N^{sym}} &= 1 - a \int_0^1 s(i) di + \frac{a^2}{2} \int_0^1 s(i)^2 di \\
&= 1 - a \cdot \frac{\varepsilon - 1}{2} V + \frac{a^2}{2} V \\
&= 1 - \frac{\varepsilon}{1 - \alpha} \cdot \frac{\varepsilon - 1}{2} V + \frac{1}{2} \left(\frac{\varepsilon}{1 - \alpha} \right)^2 V \quad (***) \\
&= 1 + \frac{\varepsilon (1 - \alpha + \alpha \varepsilon)}{2 (1 - \alpha)^2} V.
\end{aligned}$$

Price dispersion increases labor required to produce the same aggregate output.

$$\begin{aligned}
-\frac{U_N}{U_C} &= MPN = (1 - \alpha) \frac{Y}{N}, \quad \text{We need to know extra labor's utility loss.} \\
(***) \quad \left. \begin{aligned} \frac{MPN \Delta N}{Y} &= (1 - \alpha) \frac{\Delta N}{N}, \\ \frac{N - N^{sym}}{N^{sym}} &= \frac{N}{N^{sym}} - 1 = \frac{\varepsilon (1 - \alpha + \alpha \varepsilon)}{2 (1 - \alpha)^2} V \end{aligned} \right\} \Rightarrow \underbrace{\frac{MPN \Delta N}{Y}}_{\text{labor price dispersion}} = \frac{\varepsilon (1 - \alpha + \alpha \varepsilon)}{2 (1 - \alpha)} V.
\end{aligned}$$

$$\text{Define } \Theta = \frac{1 - \alpha}{1 - \alpha + \alpha \varepsilon} \quad (\text{same as lecture 8}) \Rightarrow \frac{MPN \Delta N}{Y} = \frac{1}{2} \frac{\varepsilon}{\Theta} V.$$

Now convert price dispersion to inflation. Since

$$P_t^{1-\varepsilon} = (1 - \theta) (P_t^*)^{1-\varepsilon} + \theta P_{t-1}^{1-\varepsilon},$$

log-linearizing around the zero-inflation steady state gives

$$\begin{aligned}
p_t &= (1 - \theta) p_t^* + \theta p_{t-1}, \quad p_t \equiv \log P_t. \\
\Rightarrow \pi_t &= p_t - p_{t-1} = (1 - \theta) (p_t^* - p_{t-1}) \\
&= (1 - \theta) (p_t^* - p_t) + (1 - \theta) (p_t - p_{t-1}) \\
&= (1 - \theta) (p_t^* - p_t) + (1 - \theta) \pi_t \\
\Rightarrow p_t^* - p_t &= \frac{\theta}{1 - \theta} \pi_t \Rightarrow S_t^* = \frac{\theta}{1 - \theta} \pi_t \quad \text{price-setter dispersion.}
\end{aligned}$$

$$\text{Non-price-resetters have } \begin{cases} S_t(i) = p_t(i) - p_t, \\ S_{t-1}(i) = p_{t-1}(i) - p_{t-1}, \\ p_t(i) = p_{t-1}(i). \end{cases}$$

$$\Rightarrow S_t(i) - S_{t-1}(i) = [p_t(i) - p_t] - [p_{t-1}(i) - p_{t-1}] = -p_t + p_{t-1} = -\pi_t.$$

$$\Rightarrow S_t(i) = S_{t-1}(i) - \pi_t.$$

$$\begin{aligned} V_t &= (1 - \theta) (S_t^*)^2 + \theta \int_0^1 (S_{t-1}(i) - \pi_t)^2 di \\ &= \frac{\theta^2}{1 - \theta} \pi_t^2 + \theta V_{t-1} + \theta \pi_t^2 \quad \text{We show it at the end} \\ &= \theta V_{t-1} + \frac{\theta}{1 - \theta} \pi_t^2. \end{aligned}$$

Assume $V_{-1} = 0$:

$$\begin{aligned} \sum_{t=0}^{\infty} \beta^t V_t &= \theta \sum_{t=0}^{\infty} \beta^t V_{t-1} + \frac{\theta}{1 - \theta} \sum_{t=0}^{\infty} \beta^t \pi_t^2, \\ \text{Since } V_{-1} &= 0, \quad \sum_{t=0}^{\infty} \beta^t V_{t-1} = \beta \sum_{t=1}^{\infty} \beta^{t-1} V_{t-1} + V_{-1} = \beta \sum_{t=0}^{\infty} \beta^t V_t, \\ \Rightarrow \sum_{t=0}^{\infty} \beta^t V_t &= \beta \theta \sum_{t=0}^{\infty} \beta^t V_t + \frac{\theta}{1 - \theta} \sum_{t=0}^{\infty} \beta^t \pi_t^2, \\ \Rightarrow (1 - \beta \theta) \sum_{t=0}^{\infty} \beta^t V_t &= \frac{\theta}{1 - \theta} \sum_{t=0}^{\infty} \beta^t \pi_t^2, \\ \Rightarrow \sum_{t=0}^{\infty} \beta^t V_t &= \frac{\theta}{(1 - \theta)(1 - \beta \theta)} \sum_{t=0}^{\infty} \beta^t \pi_t^2. \end{aligned} \tag{3}$$

$$\frac{MPN_t \Delta N_t}{Y_t} = \frac{1}{2} \frac{\varepsilon}{\Theta} V_t \Rightarrow \text{Total present value of welfare loss} = \frac{1}{2} \frac{\varepsilon}{\Theta} \sum_{t=0}^{\infty} \beta^t V_t.$$

Plug in (3):

$$L^\pi = \frac{1}{2} \frac{\varepsilon}{\Theta} \frac{\theta}{(1 - \theta)(1 - \beta \theta)} \sum_{t=0}^{\infty} \beta^t \pi_t^2.$$

Define

$$\lambda = \frac{(1 - \theta)(1 - \beta \theta)}{\theta} \Theta, \quad \Theta \text{ same as lecture 8.}$$

$$L^\pi = \frac{1}{2} \frac{\varepsilon}{\lambda} \sum_{t=0}^{\infty} \beta^t \pi_t^2 = \frac{1}{2} \sum_{t=0}^{\infty} \beta^t \frac{\varepsilon}{\lambda} \hat{\pi}_t^2.$$

$$L^C + L^N = \frac{1}{2} \sum_{t=0}^{\infty} \beta^t \left(\sigma + \frac{\alpha + \psi}{1 - \alpha} \right) \hat{y}_t^2.$$

Thus

$$L = L^C + L^N + L^\pi = \frac{1}{2} \sum_{t=0}^{\infty} \beta^t \left[\left(\sigma + \frac{\alpha + \psi}{1 - \alpha} \right) \hat{y}_t^2 + \frac{\varepsilon}{\lambda} \hat{\pi}_t^2 \right].$$

Why

$$\theta \int_0^1 (S_{t-1}(i) - \pi_t)^2 di = \theta V_{t-1} + \theta \pi_t^2.$$

$$\begin{aligned} \int_0^1 (S_{t-1}(i) - \pi_t)^2 di &= \int_0^1 S_{t-1}(i)^2 di + \int_0^1 \pi_t^2 di - 2\pi_t \int_0^1 S_{t-1}(i) di \\ &= V_{t-1} + \pi_t^2 - 2\pi_t \int_0^1 S_{t-1}(i) di. \end{aligned}$$

Since we have

$$\int_0^1 S_{t-1}(i) di = \frac{\varepsilon - 1}{2} V_{t-1},$$

$$\begin{aligned} \int_0^1 (S_{t-1}(i) - \pi_t)^2 di &= V_{t-1} + \pi_t^2 - 2\pi_t \cdot \frac{\varepsilon - 1}{2} V_{t-1} \\ &= V_{t-1} + \pi_t^2 - (\varepsilon - 1) V_{t-1} \pi_t. \end{aligned}$$

π_t is first-order, V_{t-1} is second-order $\Rightarrow \pi_t V_{t-1}$ is third-order.

In a second-order welfare approximation, we drop third-order terms.

$$\Rightarrow \theta \int_0^1 (S_{t-1}(i) - \pi_t)^2 di = \theta V_{t-1} + \theta \pi_t^2.$$

From

$$L = \frac{1}{2} \sum_{t=0}^{\infty} \beta^t \left[\left(\sigma + \frac{\alpha + \psi}{1 - \alpha} \right) \tilde{y}_t^2 + \frac{\varepsilon}{\lambda} \hat{\pi}_t^2 \right],$$

where

$$\lambda = \frac{(1 - \theta)(1 - \beta\theta)}{\theta} \Theta, \quad \Theta = \frac{1 - \alpha}{1 - \alpha + \alpha\varepsilon}.$$

λ is the slope of NKPC with respect to real marginal cost.

Let

$$\kappa = \lambda \left(\sigma + \frac{\psi + \alpha}{1 - \alpha} \right) \quad \dots \quad \text{slope of NKPC with respect to } \tilde{y}_t.$$

$$L = \frac{1}{2} \sum_{t=0}^{\infty} \beta^t \left[\frac{\kappa}{\lambda} \tilde{y}_t^2 + \frac{\varepsilon}{\lambda} \hat{\pi}_t^2 \right] = \frac{1}{2} \frac{\varepsilon}{\lambda} \sum_{t=0}^{\infty} \beta^t \left[\frac{\kappa}{\varepsilon} \tilde{y}_t^2 + \hat{\pi}_t^2 \right].$$

We let

$$\alpha_x = \frac{\kappa}{\varepsilon} \quad L = \frac{1}{2} \frac{\varepsilon}{\lambda} \sum_{t=0}^{\infty} \beta^t \left[\hat{\pi}_t^2 + \alpha_x \tilde{y}_t^2 \right].$$

When we have inefficiency at supply side:

$$\tilde{y}_t = \hat{y}_t - \hat{y}_t^n \neq \hat{y}_t - \hat{y}_t^e = \hat{x}_t.$$

The welfare loss is coming from the gap between realized current output $\hat{y}_t$ (deviation) and the efficient output $\hat{y}_t^e$ (deviation), thus

$$L = \frac{1}{2} \frac{\varepsilon}{\lambda} \sum_{t=0}^{\infty} \beta^t \left[\hat{\pi}_t^2 + \alpha_x \hat{x}_t^2 \right].$$

Now scale the welfare loss by $2\frac{\lambda}{\varepsilon}$:

$$L' \equiv 2\frac{\lambda}{\varepsilon}L = \sum_{t=0}^{\infty} \beta^t [\hat{\pi}_t^2 + \alpha_x \hat{x}_t^2] .$$

since $2\cdot\frac{\lambda}{\varepsilon}$ is constant, the scaling does not affect the optimal decision $(\hat{\pi}_t^*, \hat{x}_t^*)$ that minimizes the loss L' .

5.2 Taylor-type interest rate rule

$$\hat{i}_t = \phi_\pi \hat{\pi}_t + \phi_y \hat{y}_t, \quad \phi_\pi > 0, \quad \phi_y > 0.$$

$$\hat{i}_t = \phi_\pi \hat{\pi}_t + \phi_y \tilde{y}_t + \phi_y \hat{y}_t^n.$$

Plug in:

$$\begin{cases} \text{DIS:} & (\sigma + \phi_y) \tilde{y}_t + \phi_\pi \hat{\pi}_t = \sigma \mathbb{E}_t \{ \tilde{y}_{t+1} \} + \mathbb{E}_t \{ \hat{\pi}_{t+1} \} + (\hat{r}_t^n - \phi_y \hat{y}_t^n), \\ \text{NKPC:} & \hat{\pi}_t = \beta \mathbb{E}_t \{ \hat{\pi}_{t+1} \} + \kappa \tilde{y}_t. \end{cases}$$

$$\begin{bmatrix} \tilde{y}_t \\ \hat{\pi}_t \end{bmatrix} = A_T \begin{bmatrix} \mathbb{E}_t \{ \tilde{y}_{t+1} \} \\ \mathbb{E}_t \{ \hat{\pi}_{t+1} \} \end{bmatrix} + B_T (\hat{r}_t^n - \phi_y \hat{y}_t^n),$$

$$A_T = \Omega \begin{bmatrix} \sigma & 1 - \beta \phi_\pi \\ \kappa \sigma & \kappa + \beta (\sigma + \phi_y) \end{bmatrix}, \quad B_T = \Omega \begin{bmatrix} 1 \\ \kappa \end{bmatrix}, \quad \Omega = \frac{1}{\sigma + \phi_y + \kappa \phi_\pi}.$$

Assume

$$\hat{a}_t = \rho_a \hat{a}_{t-1} + \varepsilon_t^a, \quad \hat{y}_t^n = \psi_{ya}^n \hat{a}_t, \quad \psi_{ya}^n = \frac{1 + \phi}{\sigma + \phi + \alpha (1 - \sigma)} > 0.$$

$$\Rightarrow v_t = \phi_y \hat{y}_t^n = \phi_y \psi_{ya}^n \hat{a}_t.$$

$$\hat{r}_t^n = \sigma \mathbb{E}_t \{ \Delta \hat{y}_{t+1}^n \} = -\sigma \psi_{ya}^n (1 - \rho_a) \hat{a}_t.$$

$$\Delta \hat{y}_{t+1}^n = \hat{y}_{t+1}^n - \hat{y}_t^n = \psi_{ya}^n (\hat{a}_{t+1} - \hat{a}_t), \quad \mathbb{E}_t \Delta \hat{y}_{t+1}^n = \psi_{ya}^n (\rho_a - 1) \hat{a}_t.$$

$$\hat{r}_t^n - v_t = -\psi_{ya}^n [\sigma (1 - \rho_a) + \phi_y] \hat{a}_t.$$

$$B_T (\hat{r}_t^n - v_t) = \begin{bmatrix} 1 \\ \kappa \end{bmatrix} \Omega \left\{ -\psi_{ya}^n [\sigma (1 - \rho_a) + \phi_y] \right\} \hat{a}_t.$$

$$\text{Shock loading:} \quad f(\phi_y) = \frac{\sigma (1 - \rho_a) + \phi_y}{\sigma + \phi_y + \kappa \phi_\pi}, \quad f'(\phi_y) = \frac{\sigma \rho_a + \kappa \phi_\pi}{(\sigma + \phi_y + \kappa \phi_\pi)^2} > 0.$$

Shock term gets larger if the central bank responds more aggressively to output.

$\Rightarrow$ output gap and inflation variance $\uparrow$, welfare $\downarrow$.

$\phi_\pi \rightarrow +\infty \Rightarrow f(\phi_y) \rightarrow 0 \Rightarrow$ the exogenous driving force hitting the system goes to 0.

$\Rightarrow$ Aggressive inflation targeting gets arbitrarily close to the optimum.

6 A constant money growth rule: Friedman (1960)

$$\Delta \hat{m}_t = 0, \quad \forall t.$$

$$\hat{m}_t - \hat{p}_t = \hat{y}_t - \eta \hat{i}_t - \xi_t, \quad \xi_t \text{ is an exogenous money demand shock: } \Delta \xi_t = \rho_\xi \Delta \xi_{t-1} + \varepsilon_t^\xi.$$

Let

$$\widehat{\mathbb{M}}_t^+ = \widehat{\mathbb{M}}_t + \xi_t, \quad \text{denotes real balance adjusted by exogenous component of money demand.}$$

$$\widehat{\mathbb{M}}_t^+ = \tilde{y}_t + \hat{y}_t^n - \eta \hat{i}_t.$$

$$\Rightarrow \hat{i}_t = \frac{1}{\eta} \left(\tilde{y}_t + \hat{y}_t^n - \widehat{\mathbb{M}}_t^+ \right).$$

$$\widehat{\mathbb{M}}_t^+ - \widehat{\mathbb{M}}_{t-1}^+ = \hat{m}_t - \hat{m}_{t-1} - \hat{p}_t + \hat{p}_{t-1} + \xi_t - \xi_{t-1} = 0 - \hat{\pi}_t + \Delta \xi_t.$$

$$\Rightarrow \widehat{\mathbb{M}}_{t-1}^+ = \widehat{\mathbb{M}}_t^+ + \hat{\pi}_t - \Delta \xi_t.$$

$$A_{M,0} \begin{bmatrix} \tilde{y}_t \\ \hat{\pi}_t \\ \widehat{\mathbb{M}}_{t-1}^+ \end{bmatrix} = A_{M,1} \begin{bmatrix} \mathbb{E}_t \{ \tilde{y}_{t+1} \} \\ \mathbb{E}_t \{ \hat{\pi}_{t+1} \} \\ \widehat{\mathbb{M}}_t^+ \end{bmatrix} + B_M \begin{bmatrix} \hat{r}_t^n \\ \hat{y}_t^n \\ \Delta \xi_t \end{bmatrix}.$$

The degree of stability of money demand is a key element in determining the desirability of a money growth rule.